THE NATIONAL UNIVERSITY OF LESOTHO

Department of Mathematics and Computer Science

CS3520 - Computer Organisation and Architecture I

Class Test 1

2nd October 2025

1. CPU Performance

   (a) State the classic CPU performance (execution time) equation and de ne each term. Explain qualitatively how reducing each component a ects performance. [4]

   (b) Suppose a program has an instruction count (IC) of 10^9, an average CPI of 2, and runs on a 2 GHz clock. Calculate the execution time. Then calculate the new execution time if the clock frequency is doubled. [6]

2. ISA Design Tradeo s

   (a) Compare RISC and CISC instruction set designs in terms of instruction complexity and performance trade-o s. [5]

   (b) Discuss the trade-o s of xed-length versus variable-length instruction encodings. Provide one advantage and one disadvantage of each choice. [5]

3. RISC-V Code Interpretation

   (a) Consider the following RISC-V snippet. What is the nal value in register t0 ? If blt is replaced with bge t1, x0, label, what would t0 be? [5]

```
addi t0, x0, 0
addi t1, x0, 5
blt t1, x0, label
addi t0, t0, 5
label:
    addi t0, t0, 6
```

   (b) Consider the following snippet. What value is loaded into s3 ? Explain how the store and load addresses relate. [5]

```
addi t1, x0, 4
addi s2, x0, 7
sw s2, 8(t1)
lw s3, 12(x0)
```

4. RISC-V Assembly Programming

   (a) Write code to swap arr[1] and arr[3] using loads and stores. x1 contains base address of a 32-bit integer array. [5]

(b) Implement the C code below in RISC-V. Assume x in x16 and y in x17. [5]

```
if (x < y)
    y = (x - y) * 2;
else
    y = y - 1;
```

5. Translating C to RISC-V. Translate the following C function into RISC-V. Assume arr in x10 (a0), i in x11 (a1), and return in x10 (a0). [10]

```
int foo(int *arr, int i) {
    arr[i] = arr[i] + 10;
    return arr[i];
}
```

It is the work of true education to train the youth to be thinkers, and not mere re ectors of other men's thought - Education, p17